From the Fig.7.30, we see that

Volume of a sphere $= 2 \times$ (Volume of a cone)

where the diameters of sphere and cone are equal to the height of the cone.

$$= 2\left(\frac{1}{3}\pi r^2 h\right)$$

$$= \frac{2}{3}\pi r^2 \left(2r\right), \quad (\because h = 2r)$$

$$\text{Volume of a sphere} = \frac{4}{3}\pi r^3 \text{ cu. units}$$

7.3.5 Volume of a hollow sphere / spherical shell (volume of the material used)

Let r and R be the inner and outer radius of the hollow sphere.

Volume enclosed between the outer and inner spheres

$$= \frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3$$

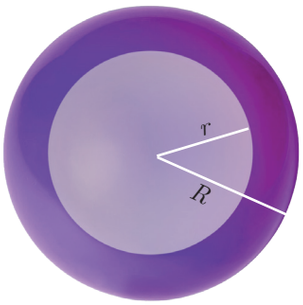


Fig. 7.31

$$\text{Volume of a hollow sphere} = \frac{4}{3}\pi(R^3 - r^3) \text{ cu. units}$$

7.3.6 Volume of solid hemisphere

Let r be the radius of the solid hemisphere.

Volume of the solid hemisphere $= \frac{1}{2}$ (volume of sphere)

$$= \frac{1}{2}\left[\frac{4}{3}\pi r^3\right]$$

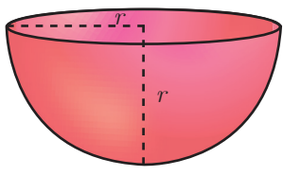


Fig. 7.32

$$\text{Volume of a solid hemisphere} = \frac{2}{3}\pi r^3 \text{ cu. units}$$

7.3.7 Volume of hollow hemisphere (volume of the material used)

Let r and R be the inner and outer radius of the hollow hemisphere.

$$\left.\begin{matrix}\text{Volume of hollow} \\ \text{hemisphere}\end{matrix}\right\} = \left[\begin{matrix}\text{Volume of outer} \\ \text{hemisphere}\end{matrix}\right] - \left[\begin{matrix}\text{Volume of inner} \\ \text{hemisphere}\end{matrix}\right]$$

$$= \frac{2}{3}\pi R^3 - \frac{2}{3}\pi r^3$$

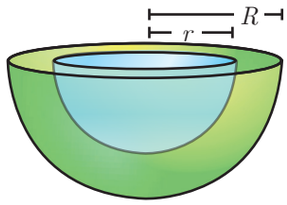


Fig. 7.33

$$\text{Volume of a hollow hemisphere} = \frac{2}{3}\pi(R^3 - r^3) \text{ cu. units}$$

Thinking Corner

A cone, a hemisphere and a cylinder have equal bases. The heights of the cone and cylinder are equal and are same as the common radius. Are they equal in volume?

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